```
MariaDB [(none)]> create database Assignment_2;
Query OK, 1 row affected (0.046 sec)

MariaDB [(none)]> use Assignment_2;
Database changed
MariaDB [Assignment_2]> create table Employee(
    -> empNo int not null primary key,
    -> fName varchar(20),
    -> lName varchar(20),
    -> address varchar(25),
    -> DOB date,
    -> sex char(1) not null,
    -> constraint check_sex check(sex in ('M','F')),
    -> position varchar(50) check (position in ('Manager', 'Team Leader',
'Analyst', 'Software Developer')),
    -> hoursWorked int check(hoursWorked >= 0 and hoursWorked <= 40));
Query OK, 0 rows affected (0.099 sec)

MariaDB [Assignment_2]> alter table Employee add deptNo int;
Query OK, 0 rows affected (0.053 sec)
Records: 0  Duplicates: 0  Warnings: 0

MariaDB [Assignment_2]> desc Employee;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| empNo       | int(11)     | NO   | PRI | NULL    |       |
| fName       | varchar(20) | YES  |     | NULL    |       |
| lName       | varchar(20) | YES  |     | NULL    |       |
| address     | varchar(25) | YES  |     | NULL    |       |
| DOB         | date        | YES  |     | NULL    |       |
| sex         | char(1)     | NO   |     | NULL    |       |
| position    | varchar(50) | YES  |     | NULL    |       |
| hoursWorked | int(11)     | YES  |     | NULL    |       |
| deptNo      | int(11)     | YES  |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
9 rows in set (0.050 sec)

MariaDB [Assignment_2]> create table Department (
    -> deptNo int not null primary key,
    -> deptName varchar(50),
    -> mgEmpNo int);
Query OK, 0 rows affected (0.007 sec)

MariaDB [Assignment_2]> describe Department;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| deptNo   | int(11)     | NO   | PRI | NULL    |       |
| deptName | varchar(50) | YES  |     | NULL    |       |
| mgEmpNo  | int(11)     | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
3 rows in set (0.031 sec)

MariaDB [Assignment_2]> create table Project(
    -> projNo int not null primary key,
    -> projName varchar(50),
    -> deptNo int);
Query OK, 0 rows affected (0.033 sec)

MariaDB [Assignment_2]> desc Project;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
```

```
| projNo   | int(11)     | NO   | PRI | NULL    |       |
| projName | varchar(50) | YES  |     | NULL    |       |
| deptNo   | int(11)     | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
3 rows in set (0.032 sec)

MariaDB [Assignment_2]> create table WorksOn(
    -> empNo int,
    -> projNo int,
    -> dateWorked date,
    -> hoursWorked decimal(5,2),
    -> primary key (empNo,projNo,dateWorked),
    -> foreign key (empNo) references Employee(empNo),
    -> foreign key (projNo) references Project(projNo));
Query OK, 0 rows affected (0.039 sec)

MariaDB [Assignment_2]> desc WorksOn;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| empNo       | int(11)      | NO   | PRI | NULL    |       |
| projNo      | int(11)      | NO   | PRI | NULL    |       |
| dateWorked  | date         | NO   | PRI | NULL    |       |
| hoursWorked | decimal(5,2) | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
4 rows in set (0.032 sec)

MariaDB [Assignment_2]> alter table department add constraint foreign key
(mgrEmpNo) references Employee(empNo);
ERROR 1072 (42000): Key column 'mgrEmpNo' doesn't exist in table
MariaDB [Assignment_2]> alter table department add constraint foreign key
(mgEmpNo) references Employee(empNo);
Query OK, 0 rows affected (0.049 sec)
Records: 0  Duplicates: 0  Warnings: 0

MariaDB [Assignment_2]> desc Department
    -> ;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| deptNo   | int(11)     | NO   | PRI | NULL    |       |
| deptName | varchar(50) | YES  |     | NULL    |       |
| mgEmpNo  | int(11)     | YES  | MUL | NULL    |       |
+----------+-------------+------+-----+---------+-------+
3 rows in set (0.029 sec)

MariaDB [Assignment_2]> alter table Project add constraint foreign key (deptNo)
references Department(deptNo);
Query OK, 0 rows affected (0.042 sec)
Records: 0  Duplicates: 0  Warnings: 0

POPULATING TABLES
a)
MariaDB [assignment_2]> INSERT INTO department (deptNo, deptName, mgEmpNo)
VALUES
    -> (10, 'Management', NULL),
    -> (20, 'Research & Development', NULL),
    -> (30, 'IT Support', NULL),
    -> (40, 'Cloud Infrastructure', NULL),
    -> (50, 'Data Analytics', NULL),
    -> (60, 'Quality Assurance', NULL),
    -> (70, 'Cybersecurity', NULL),
    -> (80, 'Product Design', NULL),
    -> (90, 'DevOps', NULL),
```

```
    -> (100, 'Human Resources', NULL);
Query OK, 10 rows affected (0.026 sec)

MariaDB [assignment_2]> UPDATE department SET mgEmpNo = 1001 WHERE deptNo = 10;
MariaDB [assignment_2]> UPDATE department SET mgEmpNo = 1002 WHERE deptNo = 20;
MariaDB [assignment_2]> UPDATE department SET mgEmpNo = 1010 WHERE deptNo = 30;
MariaDB [assignment_2]> UPDATE department SET mgEmpNo = 1011 WHERE deptNo = 40;
MariaDB [assignment_2]> UPDATE department SET mgEmpNo = 1012 WHERE deptNo = 50;
MariaDB [assignment_2]> UPDATE department SET mgEmpNo = 1013 WHERE deptNo = 60;
MariaDB [assignment_2]> UPDATE department SET mgEmpNo = 1014 WHERE deptNo = 70;
MariaDB [assignment_2]> UPDATE department SET mgEmpNo = 1015 WHERE deptNo = 80;
MariaDB [assignment_2]> UPDATE department SET mgEmpNo = 1016 WHERE deptNo = 90;
MariaDB [assignment_2]> UPDATE department SET mgEmpNo = 1017 WHERE deptNo = 100;



MariaDB [assignment_2]> SELECT * FROM department;
+--------+------------------------+---------+
| deptNo | deptName               | mgEmpNo |
+--------+------------------------+---------+
|     10 | Management             |    1001 |
|     20 | Research & Development |    1002 |
|     30 | IT Support             |    1010 |
|     40 | Cloud Infrastructure   |    1011 |
|     50 | Data Analytics         |    1012 |
|     60 | Quality Assurance      |    1013 |
|     70 | Cybersecurity          |    1014 |
|     80 | Product Design         |    1015 |
|     90 | DevOps                 |    1016 |
|    100 | Human Resources        |    1017 |
+--------+------------------------+---------+
10 rows in set (0.001 sec)

b)
MariaDB [assignment_2]> INSERT INTO project (projNo, projName, deptNo) VALUES
    -> (501, 'Enterprise ERP', 10),
    -> (502, 'NextGen AI Engine', 20),
    -> (503, 'Cloud Migration', 40),
    -> (504, 'Big Data Warehouse', 50),
    -> (505, 'Security Audit 2026', 70),
    -> (506, 'UI/UX Redesign', 80),
    -> (507, 'CI/CD Automation', 90),
    -> (508, 'Mobile App v2', 20),
    -> (509, 'Helpdesk Overhaul', 30),
    -> (510, 'Predictive Analytics', 50);
Query OK, 10 rows affected (0.025 sec)
Records: 10  Duplicates: 0  Warnings: 0

MariaDB [assignment_2]> select * from project;
+--------+----------------------+--------+
| projNo | projName             | deptNo |
+--------+----------------------+--------+
|    501 | Enterprise ERP       |     10 |
|    502 | NextGen AI Engine    |     20 |
|    503 | Cloud Migration      |     40 |
|    504 | Big Data Warehouse   |     50 |
|    505 | Security Audit 2026  |     70 |
|    506 | UI/UX Redesign       |     80 |
|    507 | CI/CD Automation     |     90 |
|    508 | Mobile App v2        |     20 |
|    509 | Helpdesk Overhaul    |     30 |
|    510 | Predictive Analytics |     50 |
+--------+----------------------+--------+
10 rows in set (0.001 sec)
```

c)

```
MariaDB [assignment_2]> INSERT INTO workson (empNo, projNo, dateWorked,
hoursWorked) VALUES
    -> (1001, 501, '2026-06-01', 10.00),
    -> (1003, 502, '2026-06-01', 40.00),
    -> (1004, 502, '2026-06-01', 35.50),
    -> (1004, 502, '2026-06-02', 25.00),
    -> (1005, 504, '2026-06-02', 38.00),
    -> (1006, 503, '2026-06-03', 40.00),
    -> (1007, 503, '2026-06-03', 20.25),
    -> (1008, 504, '2026-06-04', 30.00),
    -> (1009, 502, '2026-06-04', 15.00),
    -> (1009, 510, '2026-06-05', 12.50);
Query OK, 10 rows affected (0.002 sec)
Records: 10  Duplicates: 0  Warnings: 0

MariaDB [assignment_2]> select * from workson;
+-------+--------+------------+-------------+
| empNo | projNo | dateWorked | hoursWorked |
+-------+--------+------------+-------------+
|  1001 |    501 | 2026-06-01 |       10.00 |
|  1003 |    502 | 2026-06-01 |       40.00 |
|  1004 |    502 | 2026-06-01 |       35.50 |
|  1004 |    502 | 2026-06-02 |       25.00 |
|  1005 |    504 | 2026-06-02 |       38.00 |
|  1006 |    503 | 2026-06-03 |       40.00 |
|  1007 |    503 | 2026-06-03 |       20.25 |
|  1008 |    504 | 2026-06-04 |       30.00 |
|  1009 |    502 | 2026-06-04 |       15.00 |
|  1009 |    510 | 2026-06-05 |       12.50 |
+-------+--------+------------+-------------+
10 rows in set (0.001 sec)
```

d)

```
INSERT INTO employee (empNo, fName, lName, address, DOB, sex, position,
hoursWorked, deptNo) VALUES
    -> (1001, 'James', 'Smith', '123 Pine St, London', '1980-05-12', 'M',
'Manager', 40, 10),
    -> (1002, 'Sarah', 'Jones', '456 Oak Ave, Manchester', '1985-08-24', 'F',
'Manager', 40, 20),
    -> (1003, 'Michael', 'Brown', '789 Maple Rd, Bristol', '1988-11-02', 'M',
'Team Leader', 38, 20),
    -> (1004, 'Emily', 'Davis', '101 Cedar Ln, Leeds', '1992-03-15', 'F',
'Software Developer', 35, 20),
    -> (1005, 'David', 'Wilson', '202 Birch Dr, Liverpool', '1990-07-19', 'M',
'Analyst', 40, 50),
    -> (1006, 'Jessica', 'Taylor', '303 Ash Mews, Newcastle', '1993-01-30', 'F',
'Software Developer', 40, 40),
    -> (1007, 'Ashley', 'Thomas', '404 Elm Blvd, Sheffield', '1987-09-10', 'M',
'Team Leader', 37, 40),
    -> (1008, 'Matthew', 'White', '505 Willow Way, Bham', '1991-06-05', 'M',
'Analyst', 20, 50),
    -> (1009, 'Amanda', 'Martin', '606 Cherry Ct, Notts', '1994-12-14', 'F',
'Software Developer', 30, 20),
    -> (1010, 'Robert', 'Clark', '707 Plum La, Southampton', '1983-04-22', 'M',
'Manager', 40, 30);

MariaDB [assignment_2]> select * from employee;
+-------+---------+--------+---------------------------+------------+-----
+--------------------+-------------+--------+
| empNo | fName   | lName  | address                   | DOB        | sex |
position           | hoursWorked | deptNo |
+-------+---------+--------+---------------------------+------------+-----
```

```
+-------------------+------------+--------+
|  1001 | James   | Smith  | 123 Pine St, London      | 1980-05-12 | M   |
Manager            |          40 |     10 |
|  1002 | Sarah   | Jones  | 456 Oak Ave, Manchester  | 1985-08-24 | F   |
Manager            |          40 |     20 |
|  1003 | Michael | Brown  | 789 Maple Rd, Bristol    | 1988-11-02 | M   | Team
Leader        |          38 |     20 |
|  1004 | Emily   | Davis  | 101 Cedar Ln, Leeds      | 1992-03-15 | F   |
Software Developer |          35 |     20 |
|  1005 | David   | Wilson | 202 Birch Dr, Liverpool  | 1990-07-19 | M   |
Analyst            |          40 |     50 |
|  1006 | Jessica | Taylor | 303 Ash Mews, Newcastle  | 1993-01-30 | F   |
Software Developer |          40 |     40 |
|  1007 | Ashley  | Thomas | 404 Elm Blvd, Sheffield  | 1987-09-10 | M   | Team
Leader        |          37 |     40 |
|  1008 | Matthew | White  | 505 Willow Way, Bham     | 1991-06-05 | M   |
Analyst            |          20 |     50 |
|  1009 | Amanda  | Martin | 606 Cherry Ct, Notts     | 1994-12-14 | F   |
Software Developer |          30 |     20 |
|  1010 | Robert  | Clark  | 707 Plum La, Southampton | 1983-04-22 | M   |
Manager            |          40 |     30 |
+-------+---------+--------+--------------------------+------------+-----
+-------------------+------------+--------+
10 rows in set (0.002 sec)

2)
MariaDB [assignment_2]>  create view ProjManagedbyFemale as select
project.projNo,project.projName,project.deptNo,department.deptName,employee.fNam
e as ManagerFirstName,employee.lName as ManagerLastName from project inner join
department on project.deptNo = department.deptNo inner join employee on
department.mgEmpNo = employee.empNo where employee.sex = 'F' order by
project.projNo;
Query OK, 0 rows affected (0.041 sec)

MariaDB [assignment_2]> select * from ProjManagedbyFemale;
+--------+------------------+--------+-----------------------
+------------------+-----------------+
| projNo | projName         | deptNo | deptName               |
ManagerFirstName | ManagerLastName |
+--------+------------------+--------+-----------------------
+------------------+-----------------+
|    502 | NextGen AI Engine |     20 | Research & Development | Sarah
| Jones           |
|    508 | Mobile App v2     |     20 | Research & Development | Sarah
| Jones           |
+--------+------------------+--------+-----------------------
+------------------+-----------------+
2 rows in set (0.015 sec)

3)
a)
MariaDB [assignment_2]> CREATE VIEW EmpProject (empNo, projNo, totalHours) AS
    -> SELECT w.empNo, w.projNo, SUM(w.hoursWorked)
    -> FROM employee e, project p, workson w
    -> WHERE e.empNo = w.empNo AND p.projNo = w.projNo
    -> GROUP BY w.empNo, w.projNo;
Query OK, 0 rows affected (0.007 sec)

MariaDB [assignment_2]> select * from EmpProject;
+-------+--------+------------+
| empNo | projNo | totalHours |
+-------+--------+------------+
|  1001 |    501 |      10.00 |
|  1003 |    502 |      40.00 |
```

```
|  1004 |    502 |      60.50 |
|  1005 |    504 |      38.00 |
|  1006 |    503 |      40.00 |
|  1007 |    503 |      20.25 |
|  1008 |    504 |      30.00 |
|  1009 |    502 |      15.00 |
|  1009 |    510 |      12.50 |
+-------+--------+------------+
9 rows in set (0.006 sec)
```

b)

```
MariaDB [assignment_2]> select projNo from EmpProject where projNo = 502;
+--------+
| projNo |
+--------+
|    502 |
|    502 |
|    502 |
+--------+
3 rows in set (0.002 sec)
```

c)

```
MariaDB [assignment_2]> select count(projNo) from EmpProject where empNo = 1001;
+---------------+
| count(projNo) |
+---------------+
|             1 |
+---------------+
1 row in set (0.001 sec)
```

d)

```
MariaDB [assignment_2]> select empNo,totalHours from EmpProject group by empNo;
+-------+------------+
| empNo | totalHours |
+-------+------------+
|  1001 |      10.00 |
|  1003 |      40.00 |
|  1004 |      60.50 |
|  1005 |      38.00 |
|  1006 |      40.00 |
|  1007 |      20.25 |
|  1008 |      30.00 |
|  1009 |      15.00 |
+-------+------------+
8 rows in set (0.005 sec)
```

4)

```
MariaDB [assignment_2]> CREATE TABLE Part (
    ->     partNo INT NOT NULL,
    ->     contract VARCHAR(80) NOT NULL,
    ->     partCost INT,
    ->     PRIMARY KEY (partNo, contract)
    -> );
Query OK, 0 rows affected (0.062 sec)

MariaDB [assignment_2]> desc part;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| partNo   | int(11)     | NO   | PRI | NULL    |       |
| contract | varchar(80) | NO   | PRI | NULL    |       |
```

```
| partCost | int(11)     | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
3 rows in set (0.041 sec)

MariaDB [assignment_2]> INSERT INTO Part (partNo, contract, partCost) VALUES
    -> (101, 'Contract A', 500),
    -> (102, 'Contract B', 750),
    -> (103, 'Contract C', 1200),
    -> (104, 'Contract D', 950),
    -> (105, 'Contract E', 1500),
    -> (106, 'Contract F', 2000),
    -> (107, 'Contract G', 300),
    -> (108, 'Contract H', 1800),
    -> (109, 'Contract I', 650),
    -> (110, 'Contract J', 2500);
Query OK, 10 rows affected (0.046 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

a)

```
MariaDB [assignment_2]>  create view ExpensiveParts(partNo) as select distinct
partNo from part where partCost > 1000;
Query OK, 0 rows affected (0.034 sec)

MariaDB [assignment_2]> select * from  ExpensiveParts;
+--------+
| partNo |
+--------+
|    103 |
|    105 |
|    106 |
|    108 |
|    110 |
+--------+
5 rows in set (0.031 sec)
```

b)

ExpensiveParts can be maintained incrementally. Insertions and many updates can be handled using only the materialized view because we can determine whether a new qualifying partNo should be added. However, deletions (and updates that reduce a cost below Â£1000) generally require access to the underlying Part table because, due to the DISTINCT operator, we must determine whether another qualifying contract for the same part still exists before removing the part number from the view.